

Eduardo Alejandro Aviles Jimenez

✉ eduardo.avilesj@gmail.com | ☎ (+502) 3996-5277 | in eduardoavilesj | 🌐 steelcolusus | 📍 Guatemala

Backend & Machine Learning Engineer | Python, Data Pipelines & AWS

Engineer with 10+ years of experience across backend development, data engineering, and production AI/ML systems in financial services, social media, and enterprise applications. Owns data and model architecture from ingestion and normalization through training, inference, and monitoring on AWS. Experienced in microservices, multi-source ETL, cloud infrastructure, and CI/CD.

WORK EXPERIENCE

LoanPro — *Machine Learning Engineer* JUN 2022 - PRESENT

- Built production data and model pipelines with **Python**, **Pandas**, and **scikit-learn** for loan monitoring and anomaly detection, translating servicing data and analyst needs into deployed systems.
- Led data and model architecture across risk intelligence, anomaly detection, retrieval pipelines, and conversational workflows, balancing explainability and reliability in financial services.
- Established reusable OpenSearch/API normalization pipelines for nested loan, payment, and servicing records, improving model-readiness across multiple AI systems.
- Built repeatable training, inference, validation, and monitoring workflows with **MLflow** and **AWS ECS/Fargate** to support production model reliability.
- Reduced noisy monitoring events by 25% through anomaly detection and tenant-specific thresholding, improving operational signal quality.
- Delivered portfolio-risk and monitoring products aligned with IFRS 9/ECL workflows, including retrieval-grounded analyst tools and explainable recommendations.

BairesDev - HipTrain — *Senior Software Engineer* AUG 2021 - JUN 2022

- Led backend, infrastructure, and CI/CD architecture for training and scheduling platforms, improving deployment times by 30%.
- Developed core scheduling and reservation systems for training classes, improving scheduling efficiency by 80%.
- Implemented Amazon Lex chatbot workflows that automated 40% of customer inquiries and improved user interaction efficiency.

LogMeIn — *BI/ML Developer* APR 2016 - DEC 2021

- Architected multi-source data integration and ETL microservices, including live-data streaming, improving scalability and processing reliability.
- Implemented ML algorithms for product enhancements, improving prediction accuracy by 20%.

SSNAPP (Social Media Gateways) — *Senior Software Developer* SEP 2014 - APR 2016

- Led transition from monolithic architecture to microservices using Netflix OSS, increasing system scalability by 50% and reducing downtime by 20%.
- Optimized back-end infrastructure and implemented AWS solutions, reducing infrastructure costs by 30%.

SKILLS

Backend & APIs	Python · FastAPI · Django · Flask · SQL · Java · Spring Boot · Node.js · NestJS · TypeScript
Data & Pipelines	Pandas · ETL pipelines · API integrations · PostgreSQL · OpenSearch · Redis · DynamoDB · Redshift
ML & MLOps	Scikit-learn · XGBoost · TensorFlow · Keras · MLflow · DVC · Weights & Biases · model monitoring
Cloud & DevOps	AWS ECS · Fargate · Lambda · Docker · Kubernetes · Terraform · CloudFormation · Ansible · Jenkins · GitHub Actions
LLM & Retrieval	LlamaIndex · LangChain · Ollama · Chainlit · Pydantic

SELECTED DATA / ML PROJECTS

Loan Monitor — *Loan creation anomaly monitoring*

LOANPRO · 2024

Designed and deployed a loan-creation monitoring system that clusters user behavior and flags anomalies; supported per-tenant targeting with repeatable training and inference on AWS ECS/Fargate.

Python Pandas MLflow Scikit-learn ECS Fargate

Risk Radar — *Loan delinquency prediction system*

LOANPRO · JAN 2025 – PRESENT

Built probability-weighted delinquency models across Current, Early/Severe Delinquency, and Default stages under IFRS 9/ECL; shipped tenant risk leaderboards, drill-down analysis, and early-warning alerts for portfolio managers and servicing teams.

XGBoost Python Pandas IFRS 9/ECL MLflow

ML Pipeline Studio — *Churn, Rental Pricing, and Dynamic-Risk Pipelines*

PERSONAL · 2024

Built modular ML pipelines for churn prediction, automated weekly rental-price retraining, and full lifecycle management across ingestion, scoring, retraining, and deployment.

Python MLflow CI/CD model monitoring

Loan Health Intelligence — *AI-powered credit health assessment platform*

LOANPRO · FEB – APR 2025

Delivered an AI loan-risk platform with structured credit health assessments, OpenSearch and API retrieval, schema-enforced outputs, and explainable recommendations.

LlamaIndex Ollama OpenSearch Chainlit Pydantic

Search Analytics Copilot — *Conversational OpenSearch analytics assistant*

LOANPRO · MAY – AUG 2025

Built function-calling assistants for index discovery, metadata inspection, JSON querying, visual trend generation, loan-context retrieval, and API fallback.

LangChain LlamaIndex Chainlit OpenSearch Matplotlib

AWARDS

First Place · LoanPro Hackathon — *Risk Radar*

JAN 2025

First Place · LoanPro Hackathon — *Loan Monitoring*

APR 2024

First Place & Developers' Choice · LogMeIn Hackathon — *LSTM churn prediction*

JUL 2018

CERTIFICATIONS – UDACITY NANODEGREES

Machine Learning DevOps Engineer

CI/CD · FastAPI · MLflow · Weights & Biases

Cloud DevOps Engineer

CloudFormation · Ansible · Jenkins · Kubernetes

Artificial Intelligence

Search · planning · pattern recognition

Deep Learning

CNNs · RNNs · GANs · sentiment analysis

EDUCATION

Computer Systems Engineering — *Engineering Degree*

JAN 2005 - SEPT 2011

Universidad De San Carlos De Guatemala

SPEAKING ENGAGEMENTS

Neural Networks from Scratch · Facebook Dev Circles Guatemala

MAR 2019

Building a working network in raw Python with no ML libraries.

RNNs and Sequence-to-Sequence Models · Facebook Dev Circles Guatemala

OCT 2018

Generating social-media threads from a Spanish-language dataset.

Neural Machine Translation · Mayan Linguistic Conference, Guatemala

JUL 2018

Applying NMT concepts toward Mayan dialect preservation.

Generative Adversarial Networks · Google IO Extended Guatemala (keynote)

MAY 2018

Facial image generation and applied generative modeling.